Silica concentrate prediction in mining process

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INTRODUCTION

- Iron ores are rocks and minerals from which metallic iron can be economically extracted.
- In steel manufacturing industries, the high-grade iron ores plays a vital role in the steel manufacturing process.
- If the impurities are high in iron ores, the efficiency and performance of the kiln are reduced.
- The common impurity present in the Iron ore is silica.
- The typical magnetite iron-ore concentrate contains impurity of 3–7% of silica.
- Therefore predicting the percentage of silica concentrate present in the Iron ore is essential.

DATASET DESCRIPTION

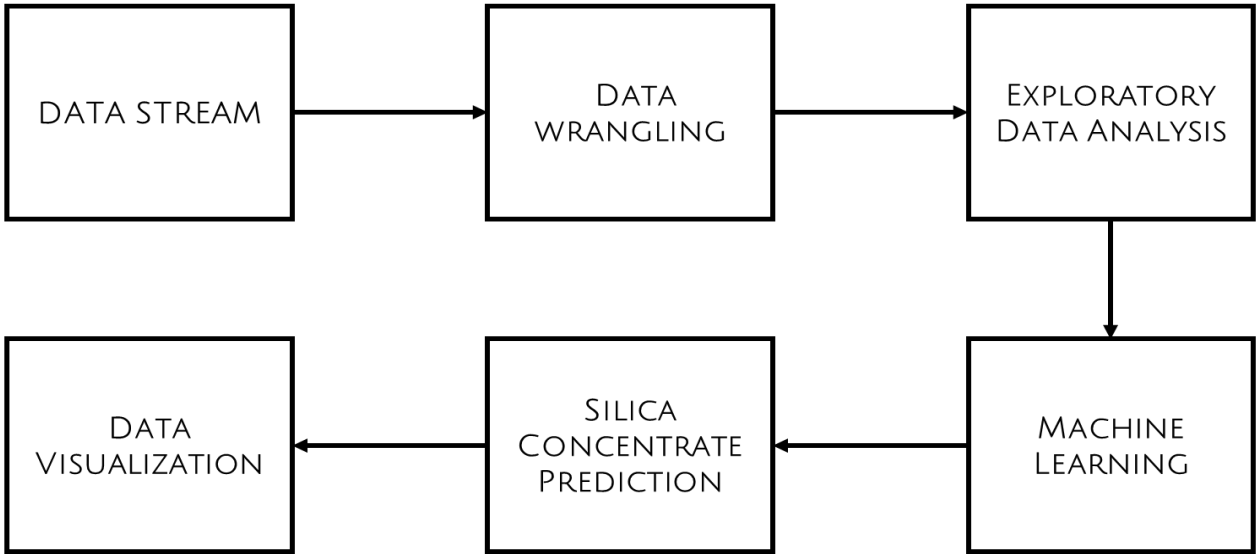
- Duration : 6 months (March - September 2017)
- Measurements are taken for every 20 seconds
- The dataset has **7,37,453** samples and **24** attributes

Dataset Description (cont.)

- **Independent Variables** : *['% Iron Feed', '% Silica Feed', 'Starch Flow', 'Amina Flow', 'Ore Pulp Flow', 'Ore Pulp pH', 'Ore Pulp Density', 'Flotation Column 01 Air Flow', 'Flotation Column 02 Air Flow', 'Flotation Column 03 Air Flow', 'Flotation Column 04 Air Flow', 'Flotation Column 05 Air Flow', 'Flotation Column 06 Air Flow', 'Flotation Column 07 Air Flow', 'Flotation Column 01 Level', 'Flotation Column 02 Level', 'Flotation Column 03 Level', 'Flotation Column 04 Level', 'Flotation Column 05 Level', 'Flotation Column 06 Level', 'Flotation Column 07 Level', '% Iron Concentrate']*
- **Dependent Variable** : *[' % Silica Concentrate']*

Overview of Data Analytics

Silica Concentrate Prediction in Mining Process



EXISTING WORK

- The percentage of Iron and silica concentrate is predicted.
- Multi-Target Regression is used (% Iron and Silica concentrate)
- Machine Learning Algorithms like ADAboost, Random Forest, Decision tree, XGboost, Ridge are used.

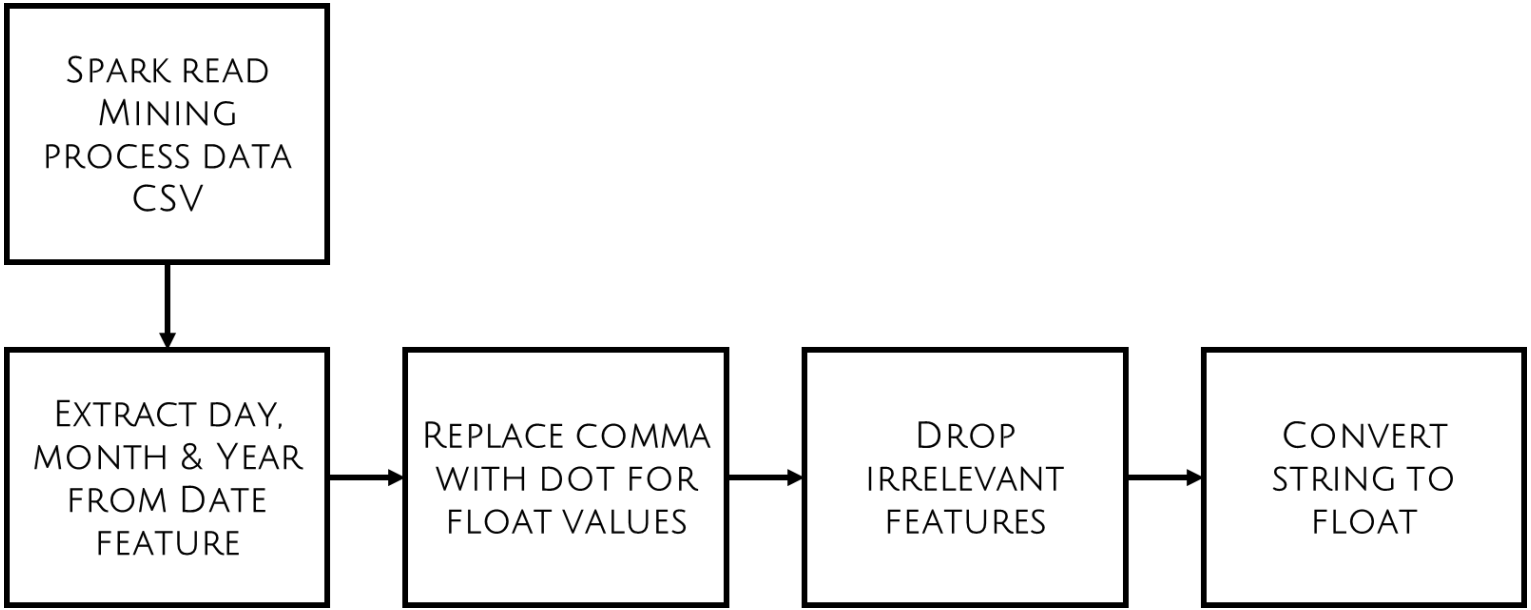
MODEL	R2 Score	RMSE
ADA BOOST	0.9745449	0.0320345
RANDOM FOREST	0.9404630	0.0749937
DECISION TREE	0.9036198	0.1243733
XBOOST	0.7750269	0.2641687
RIDGE	0.1457070	1.0722436

OBJECTIVE

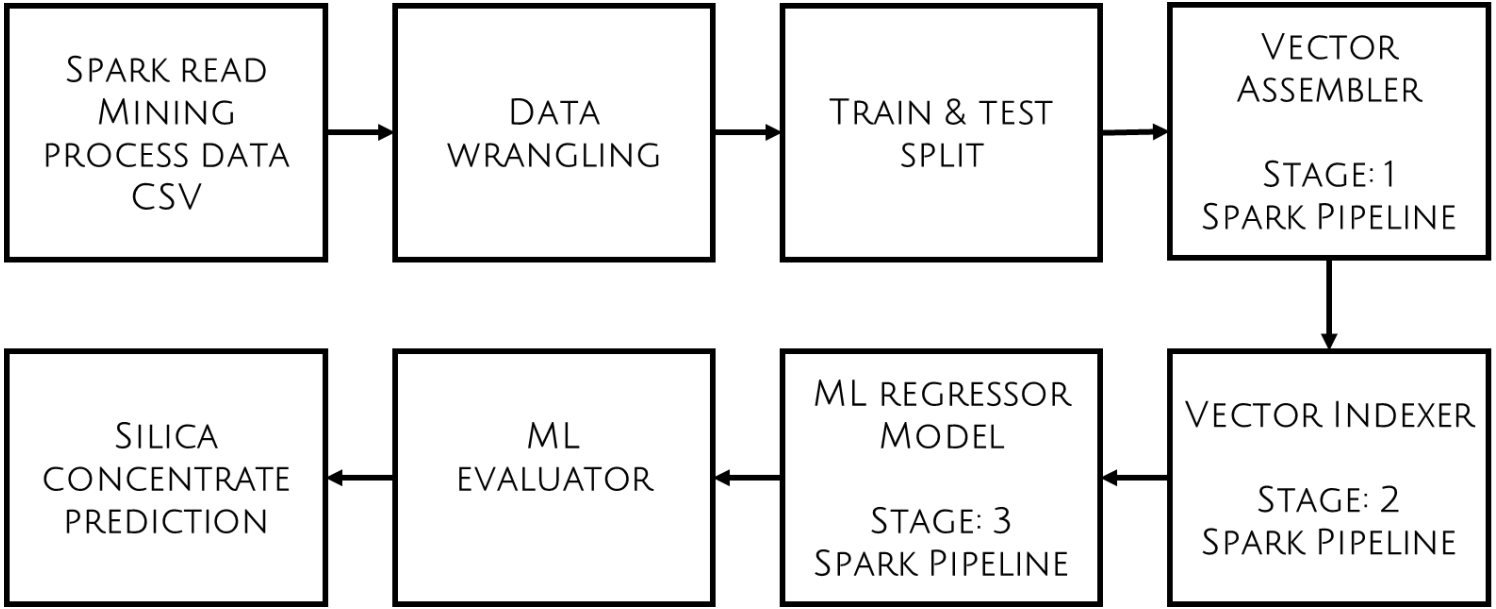
To predict the percentage of silica concentrate from Mining process database

- ❑ To find a suitable Machine Learning algorithm
- ❑ To stream real-time data and perform prediction
- ❑ To deploy the solution in Google Cloud Platform

Data Preprocessing



Prediction using Pyspark



Prediction using PySpark (cont.)

```
train, test = df.randomSplit([0.7, 0.3])
```

Train & test split

```
featuresCols = df.columns
featuresCols.remove('silica_con_pct')
vectorAssembler = VectorAssembler(inputCols=featuresCols, outputCol="rawFeatures")
vectorIndexer = VectorIndexer(inputCol="rawFeatures", outputCol="features", maxCategories=4)
```

Vector Assembler & Vector Indexer

SPARK ML

Machine Learning Algorithms used

- ❑ Linear Regression
- ❑ Decision Tree
- ❑ Random Forest
- ❑ Gradient Boost Trees

ML : RESULTS

```
[ ]  1 import pandas as pd
     2 r2_arr = [lr_r2Score,dt_r2Score, rf_r2Score, gbt_r2Score]
     3 rmse_arr = [lr_rmse,dt_rmse, rf_rmse, gbt_rmse]
     4 reg_arr = ['Linear Regression', 'Decision Tree', 'Random Forest', 'Gradient Boosting Trees']
     5 pd.DataFrame({'Regressor': reg_arr,'RMSE': rmse_arr, 'R2_Score': r2_arr})
```

	Regressor	RMSE	R2_Score
0	Linear Regression	0.628834	0.687272
1	Decision Tree	0.074738	0.995582
2	Random Forest	0.248975	0.950976
3	Gradient Boosting Trees	0.100799	0.991965

Comparison of Evaluation Metrics

Spark - Streaming

```
ssc = StreamingContext(spark.sparkContext, batchDuration= 10)
lines = ssc.socketTextStream("127.0.0.1", 9991)
words = lines.flatMap(lambda line : line.split("MINING_DATA"))
words.foreachRDD(newRecordPrediction)
ssc.start()
ssc.awaitTermination()
```

Pyspark Streaming

SPARK – STREAMING (CONT.)

```
(base) charitha@ubuntu:~/streaming/ml-streaming$ nc -lk 9991
2017-04-03 14:00:00,"54,38","17,41","2424,02","567,017","392,578","9,52567","1,78265","250,488","248,95","249,39","295,09
6","306,4","249,829","249,829","594,08","756,146263587","883,9220341686","566,797","457,699","544,937","516,445","62,05",
"4,142098834"
```

Netcat

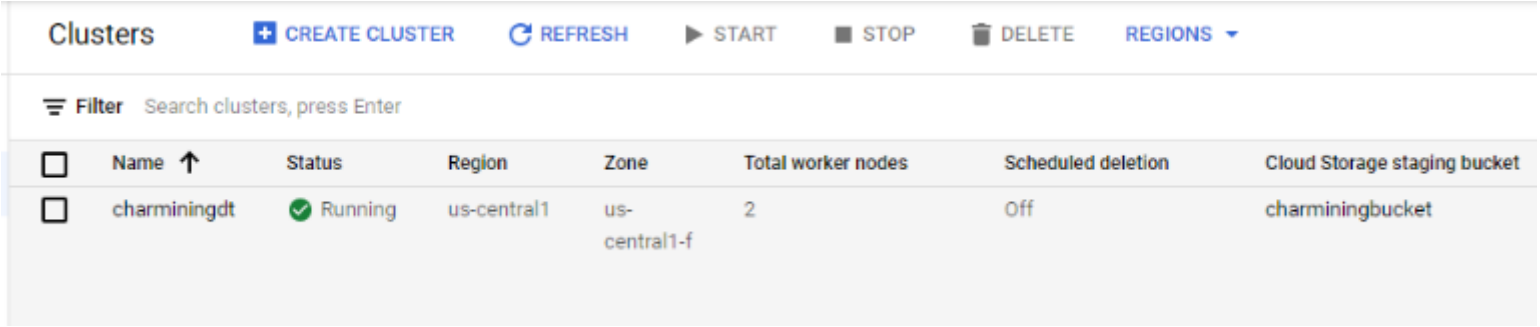
```
predicting % silica concentrate...
+--------------+------------------+
|silica_con_pct|        prediction|
+--------------+------------------+
|     1.5877402|1.5853303670883179|
|     1.5853304|1.5853303670883179|
|     1.5829206|1.5829205513000488|
|     1.5805109|1.5925596952438354|
+--------------+------------------+



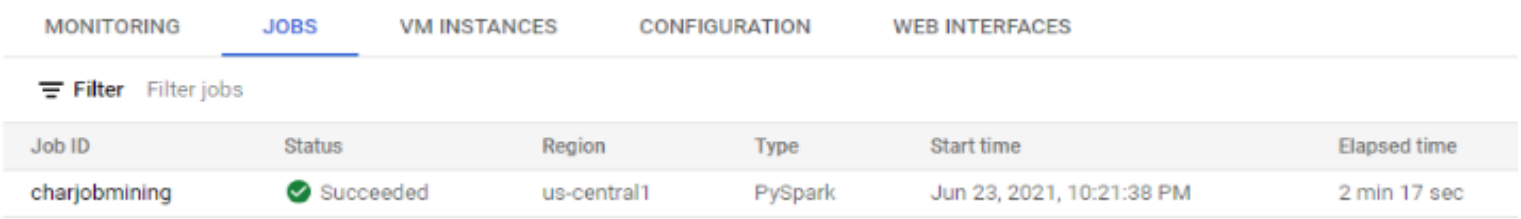
message received...
No data RDD is empty
```

Streaming - Prediction Results

DATAPROC



Cluster



Job

DATAPROC (CONT.)

Filter by name prefix only | Filter | Filter objects and folders

Name	Size	Type	Created time	Storage class	Last modified	Public access
dt_mining_data_proc.py	5.5 KB	application/octet-stream	Jun 23, 2021, ...	Standard	Jun 23, 20...	Not public
nano_data.csv	11.9 MB	application/vnd.ms-excel	Jun 23, 2021, ...	Standard	Jun 23, 20...	Not public

Files in the bucket

MONITORING | JOBS | VM INSTANCES | CONFIGURATION | WEB INTERFACES

Filter | Filter instances

Name	Role
charminingdt-m	Master
charminingdt-w-0	Worker
charminingdt-w-1	Worker

Master and Worker Nodes

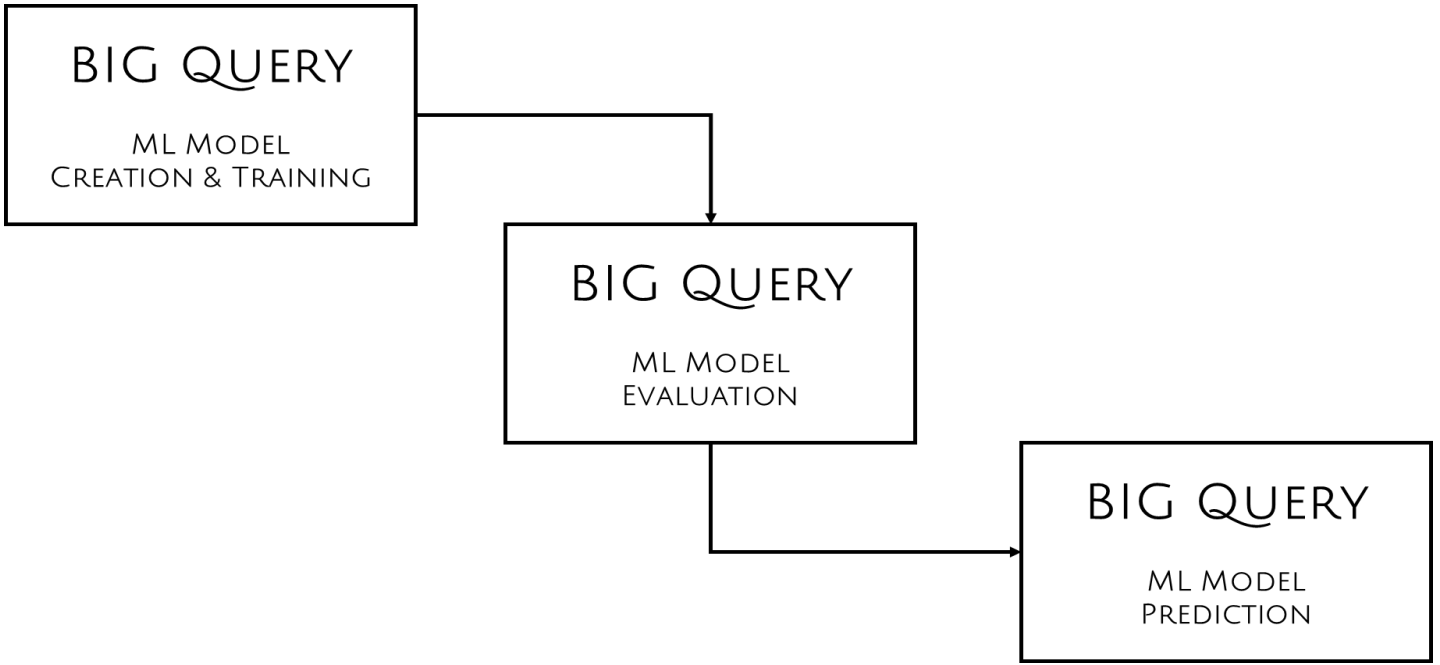
DATAPROC (CONT.)

Decision Tree Regressor

```
RMSE:  0.31337340324715307
R2 Score:  0.923465239087214
21/06/23 16:53:50 WARN org.apache.spark.scheduler.DAGScheduler: Broadcasting large task binary with size 1767.4 KiB
+--------------+------------------+
|silica_con_pct|        prediction|
+--------------+------------------+
|          2.76| 2.759999990463257|
|          2.76| 2.759999990463257|
|          2.76| 2.759999990463257|
|          1.73|1.7300000190734863|
|          1.73|1.7300000190734863|
+--------------+------------------+
only showing top 5 rows
```

Decision Tree dataproc Prediction

Big Query


BIG QUERY
ML MODEL
CREATION & TRAINING
BIG QUERY
ML MODEL
EVALUATION
BIG QUERY
ML MODEL
PREDICTION

Big Query - Results

Job information | Results | JSON | Execution details

Row	mean_absolute_error	mean_squared_error	mean_squared_log_error	median_absolute_error	r2_score	explained_variance
1	0.48476098674712176	0.398561534977726	0.03494336781305392	0.3843717631377024	0.6850739018292089	0.6850744137834837

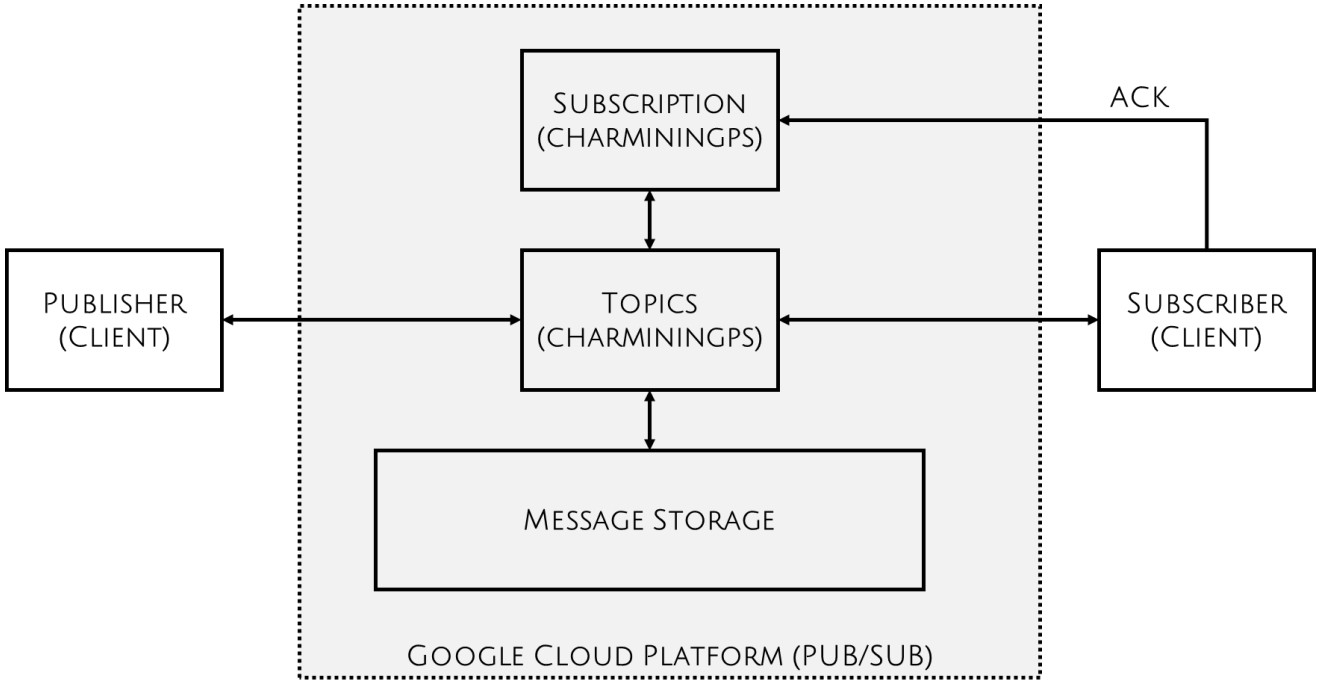
Evaluation Metrics

Job information | Results | JSON | Execution details

Row	predicted___Silica_Concentrate	__Silica_Concentrate
1	3.169117970353104	3.17

Prediction of % silica concentrate

Pub/Sub


Subscription
(charminingps)
ACK
Publisher
(Client)
Topics
(charminingps)
Subscriber
(Client)
Message Storage
Google Cloud Platform (pub/sub)

PUB/SUB (CONT.)

Installations

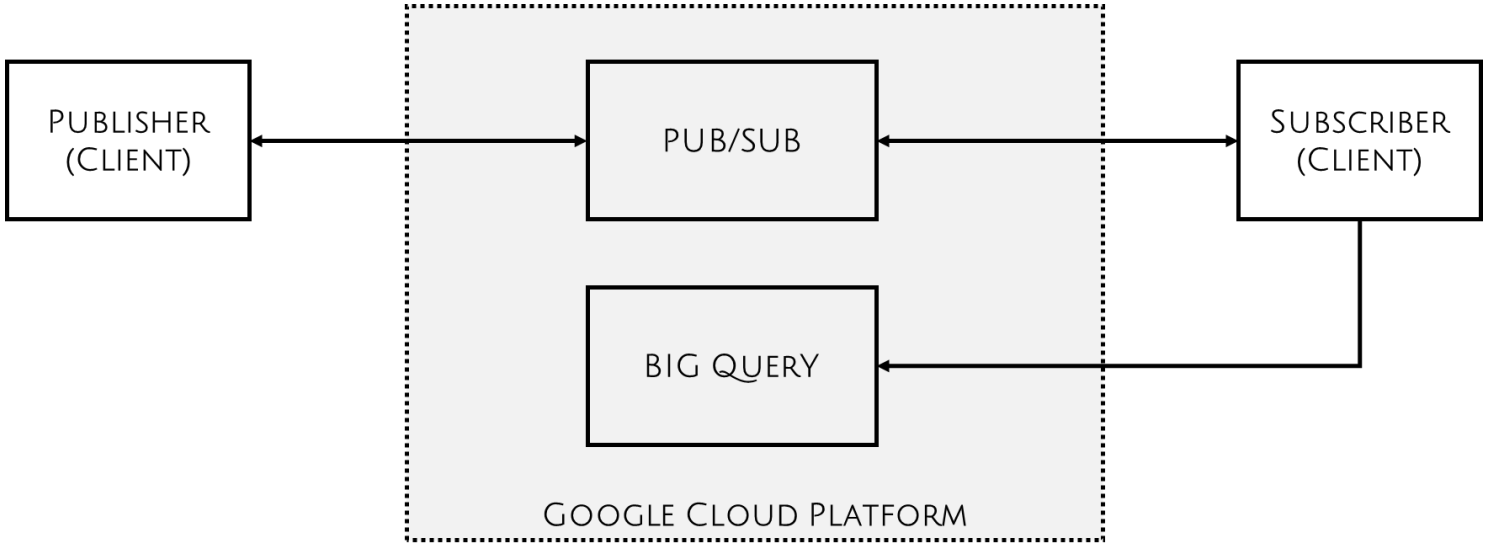
- Virtual Environment
- Google Cloud Pub/Sub library
- Google Cloud Big query library

PUB/SUB (CONT.)

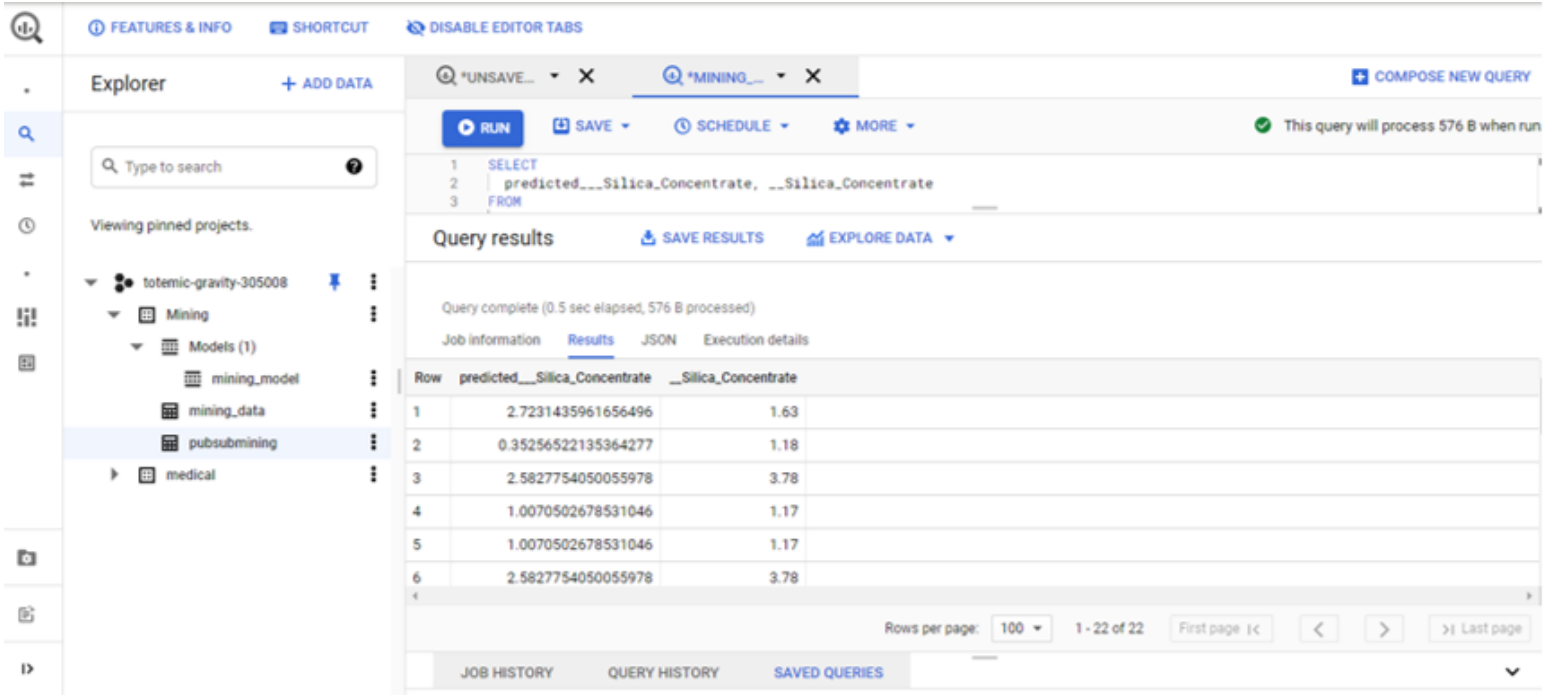
Publish time	Attribute keys	Message body	Ack
Jun 24, 2021, 11:37:58 PM	—	Mining data	Deadline exceeded
Jun 25, 2021, 12:11:27 AM	—	date,% Iron Feed,% Silica Feed,Starch Flow,Amina Flow,Ore Pulp Flow,Ore Pulp Pulp Density,Flotation Column 01 Air Flow,Flotation Column 02 Air Flow,Flotation Column 03 Air Flow,Flotation Column 04 Air Flow,Flotation Column 05 Air Flow,Flotation Column 06 Air Flow,Flotation Column 07 Air Flow,Flotation Column 01 Level,Flotation Column 02 Level,Flotation Column 03 Level,Flotation Column 04 Level,Flotation Column 05 Level,Flotation Column 06 Level,Flotation Column 07 Level,% Iron Concentrate,% Silica Concentrate	Deadline exceeded
Jun 25, 2021, 12:11:28 AM	—	2017-05-28 04:00:00,"64,03","6,26","3114,33","402,284","399,452","9,77276","1,60066","300,449","298,377","299,78","301,082","298,935","299,994","591,924","603,15","596,868","506,789","413,855","393,361","66,42","1,17"	Deadline exceeded
Jun 25, 2021, 12:11:28 AM	—	2017-07-24 13:00:00,"57,46","10,8","3729,41","581,572","400,691","9,80857","1,68255","250,261","249,213","294,3610752454","293,9049640131","255,81","248,773","386,377","401,755","528,939","526,975","505,296","469,285","64,76","3,78"	Deadline exceeded
Jun 25, 2021, 12:11:29 AM	—	2017-06-09 06:00:00,"64,03","6,26","4530,01","409,609","403,781","10,2608","1,67864","299,269","301,55","299,707","300,34","300,304","301,127","406,136","436,271","393,695","388,358","393,525","419,612","65,8","1,47"	Deadline exceeded
Jun 25, 2021, 12:11:30 AM	—	2017-04-13 22:00:00,"57,54","11,23","2080,37","547,956","405,026","9,87116","1,77447","251,756","248,773","295,6592478006","299,0336046595","249,128","253,044","598,039","601,93","455,695","448,587","592,779","602,627","66,46","1,29"	Deadline exceeded

Verification of the published Messages on GCP-subscriptions

Big Query

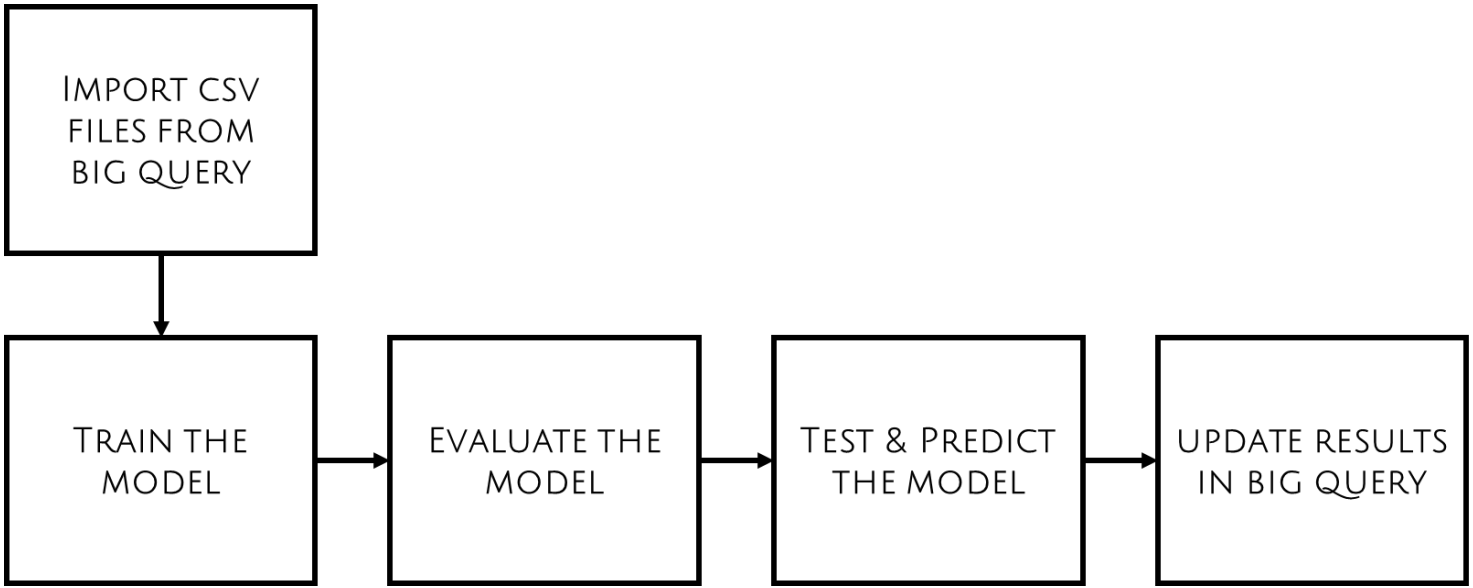

Publisher
(Client)
Pub/Sub
Subscriber
(Client)
Big Query
Google Cloud Platform

Big Query (Cont.)

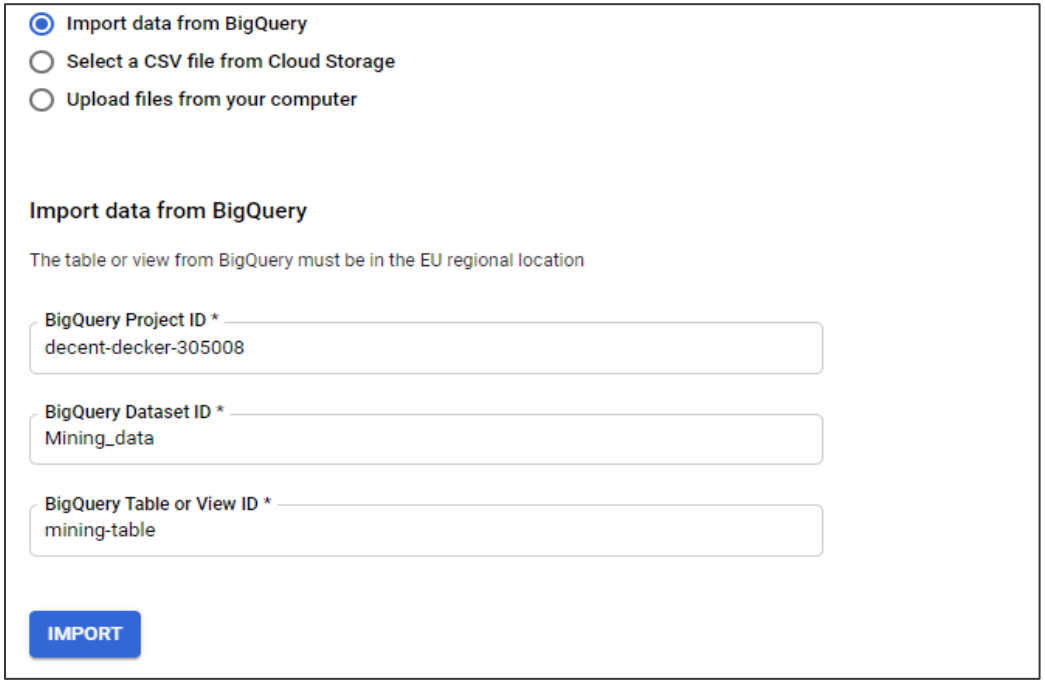


Predicted results of newly created big query table

AUTOML

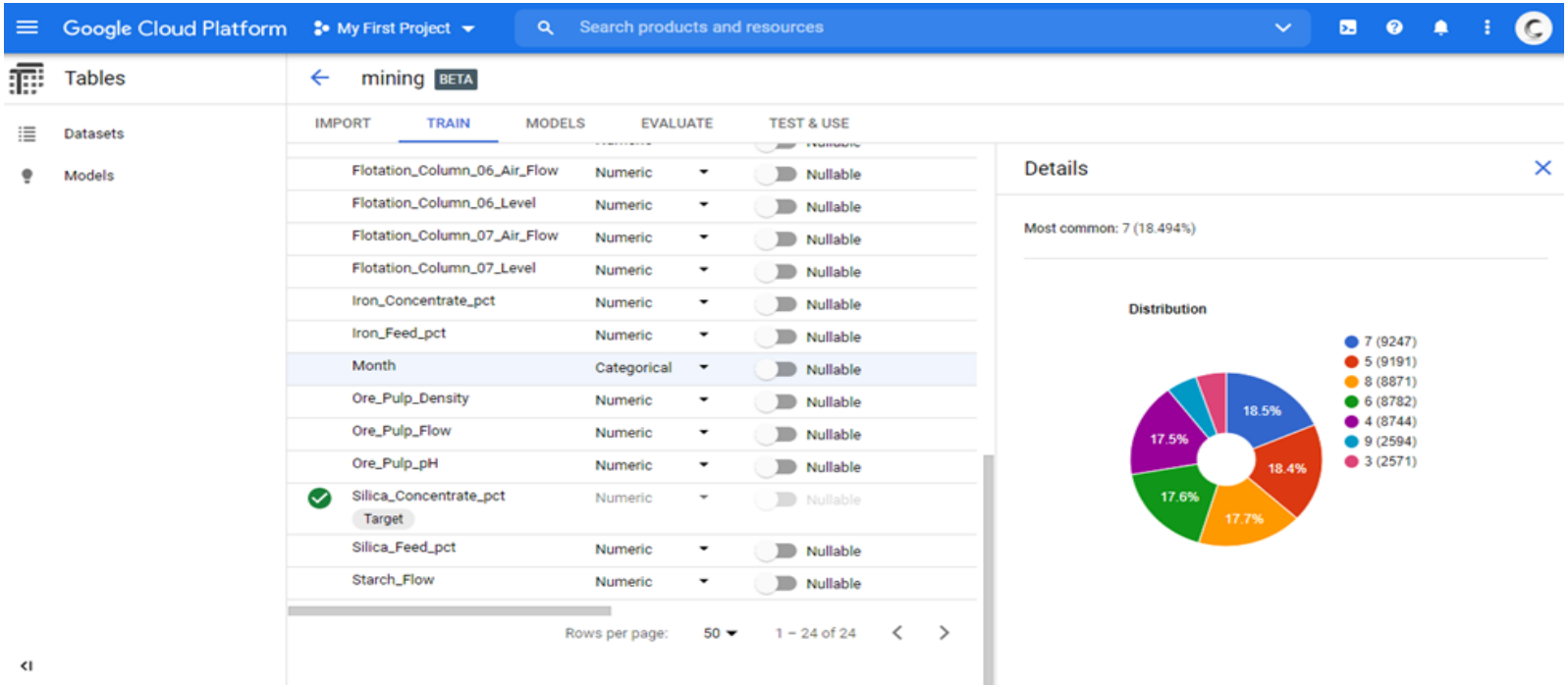

Import csv files from big query
Train the model
Evaluate the model
Test & Predict the model
update results in big query

AUTOML (CONT.)



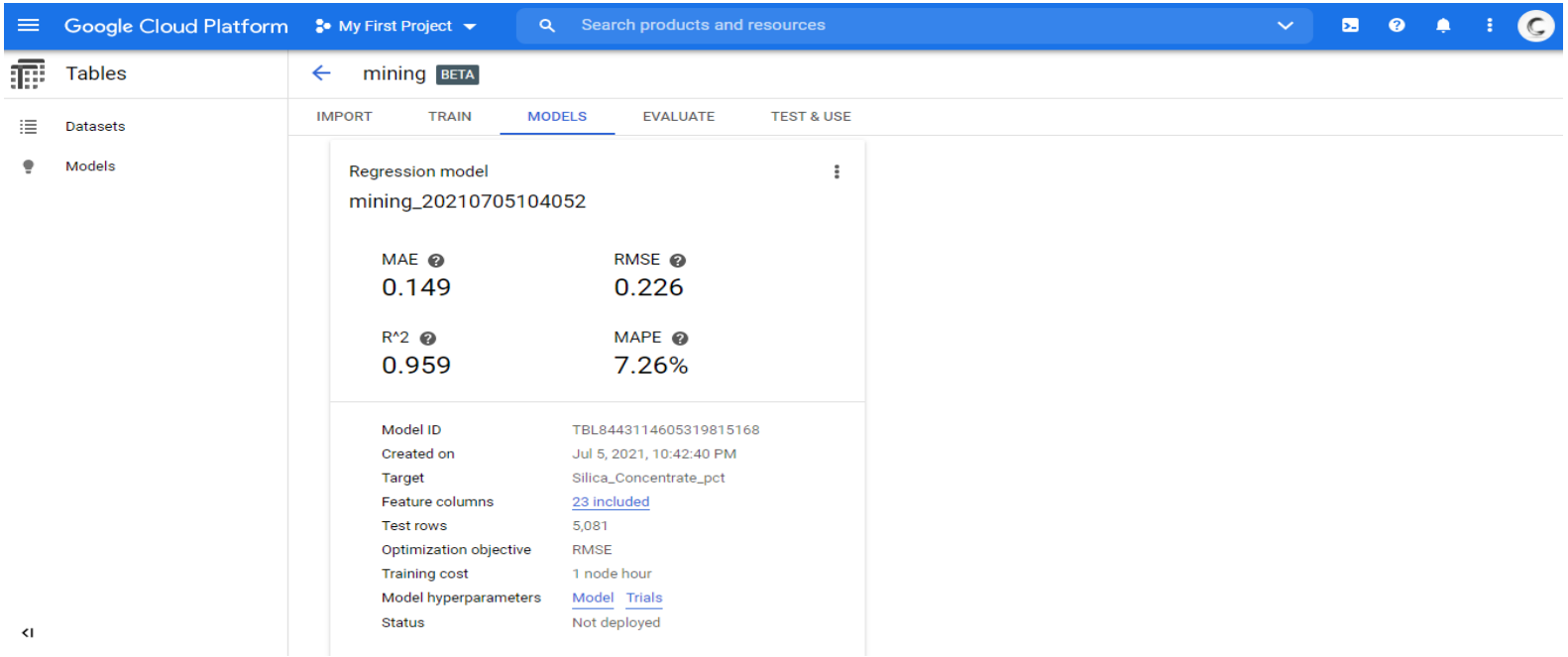
Importing data from Big query

AUTOML (CONT.)



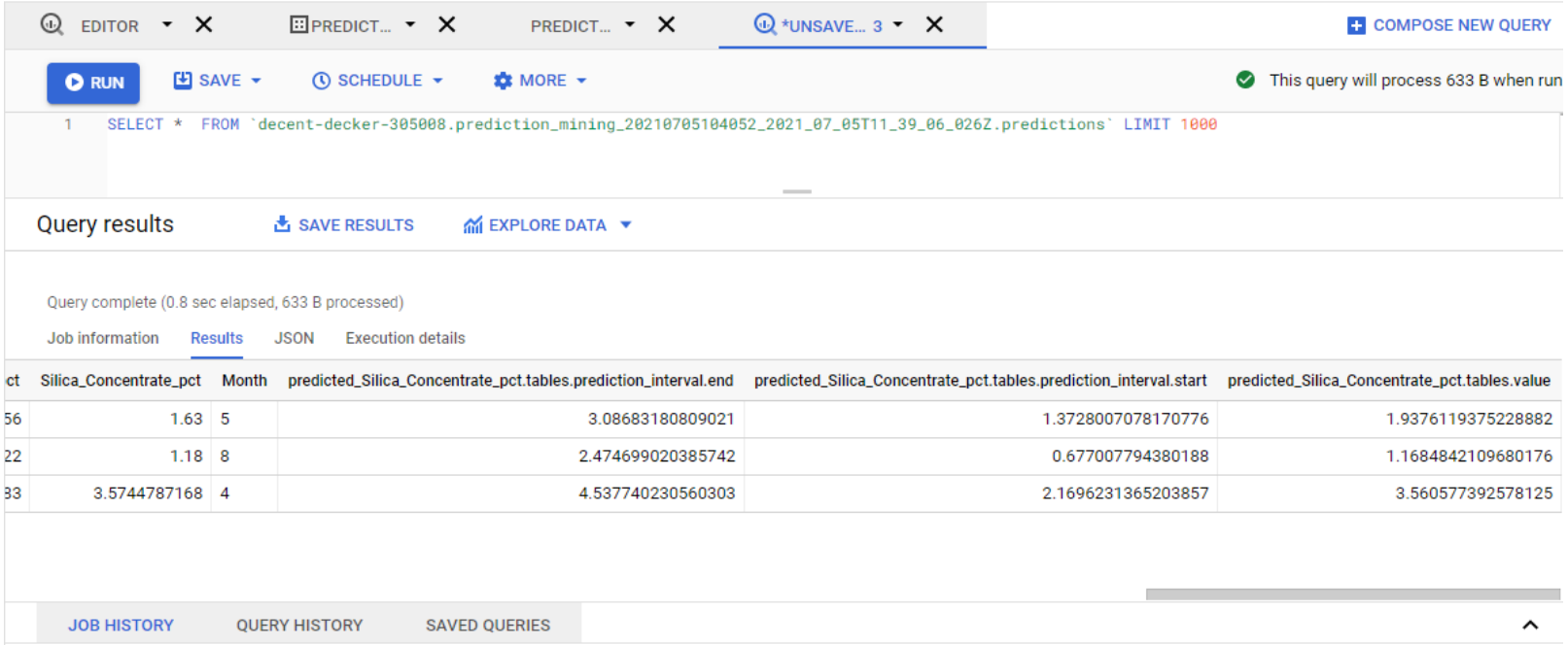
Distribution of data

AUTOML (CONT.)



Training Model

AUTOML (CONT.)



ct	Silica_Concentrate_pct	Month	predicted_Silica_Concentrate_pct.tables.prediction_interval.end	predicted_Silica_Concentrate_pct.tables.prediction_interval.start	predicted_Silica_Concentrate_pct.tables.value
56	1.63	5	3.08683180809021	1.3728007078170776	1.9376119375228882
22	1.18	8	2.474699020385742	0.677007794380188	1.1684842109680176
33	3.5744787168	4	4.537740230560303	2.1696231365203857	3.560577392578125

Predicted Results in Big query

Thank You